Who's In Charge?: A Discrete Simulation of Bank Efficiency

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***Abstract*—This report presents the development and evaluation of a Discrete Event Simulation (DES) system to model customer flow and teller utilization in a banking environment. By simulating various queueing strategies, teller configurations, and customer arrival patterns, the project aims to identify approaches that optimize throughput and minimize customer wait times. Key experiments include comparisons between FIFO and priority-based queues, as well as system performance under different numbers of tellers. Results suggest that adding tellers substantially increases throughput and reduces wait times, while prioritizing light requests has marginal effects. Conversely, non-uniform arrival patterns significantly degrade performance, offering crucial insights for real-world scheduling and resource allocation.**

Keywords— First in First Out (FIFO), priority-queue (PQ), Discrete Event Simulation (DES), Work Units (WUs)

I. Motivation

Efficient customer service is critical to operational success in many service-based industries, including banking. Long wait times and poor throughput not only impact customer satisfaction but also affect operational efficiency and profitability. This simulation explores the impact of queueing policies and resource allocation to discover ways to improve system performance with minimal changes.

II. Problem Statement

The goal of this simulation is to investigate how queueing strategies, teller allocation, and customer arrival patterns affect key performance indicators: average wait time, number of customers served/unserved, and teller utilization. The overarching question is: How can we increase customer throughput and service quality in a simulated banking environment?

III. Related Work and Background material

Variations in scheduling strategies and how service resources are allocated can significantly impact the performance of customer service systems. While FIFO (First-In-First-Out) queues are frequently used, they may not always provide the best service outcomes. Adjusting the number of service windows and giving preference to certain types of customer requests can influence both wait times and the total number of customers served.

IV. Approach

The simulation modeled an 8-hour bank workday with a configurable number of service windows. The base case used 10 windows, each processing 10 work units (WUs) per hour. Scenarios with 9 and 11 windows were also tested to observe effects of under- and over-provisioning. Customer arrivals followed either a uniform distribution or a bell-curve centered within the day. Each customer's service demand was sampled from a truncated normal distribution with a mean of 10, standard deviation of 2, and bounds of 5 to 15 WUs. Three queuing strategies were tested: a standard FIFO model, a priority queue that reserved one window for light requests (5.0–5.5 WUs), and configurations with altered window counts. Metrics collected included average wait time, customers served/unserved, and teller utilization.

V. Experiment Set-up (including Metrics, Benchmarks, Data Collection)

The simulation ran 10,000 iterations for each configuration to ensure statistical significance. Metrics included average customer wait time, number of customers served, number of customers unserved, and utilization rate of service windows. Benchmarks were established using the base FIFO model with 10 service windows and uniform arrivals. Data was collected and averaged across runs, providing consistent results for comparative analysis.

TABLE I. Summary of Simulation Results

Scenario	*Avg. Wait Time (min)*	*Customers Served*	*Customers Unserved*	*Utilization*
Base Case (10 windows, FIFO)	17.49	137	23	92.81
Priority Queue (10 windows)	30.67	129	31	87.47
Extra Window (11 windows, FIFO)	5.40	146	14	89.85
Fewer Windows (9 windows, FIFO)	43.63	125	35	94.14
Non-Uniform Arrivals (10 windows)	228.09	68	92	88.85

Fig. 1. Table summarizing the impact of different queue strategies, window configurations, and customer arrival patterns on service efficiency, customer throughput, and wait times across simulated bank scenarios.

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VI. Results and Discussion

The configuration with an extra window (11 total) yielded the highest throughput, serving 146 customers with an average wait time of just 5.40 minutes. In contrast, the fewest windows cases (9 total) resulted in the longest wait times, 43.63 minutes on average, and the lowest throughput. Notably, the non-uniform arrival case, despite having the same number of windows as the base case, showed poor performance due to traffic peaking mid-day, overwhelming the system and leading to only 68 customers served. These findings highlight the importance of aligning service capacity with expected traffic patterns and suggest that modest increases in resources can lead to significant performance improvements.

VII. Contributions and Conclusions

This study contributes to the understanding of how service system configurations impact customer flow and resource efficiency. By systematically altering queue strategies and service window availability, we identified that adding even a single service window drastically improves outcomes, while unbalanced arrival distributions can cripple throughput. These insights are directly applicable to real-world systems, providing guidance for resource allocation and policy design in service environments. The project demonstrates that DES is an effective tool for testing operational hypotheses before implementation.

VII. Meta-Presentation Analysis

The delivery of the simulation results through the presentation was overall clear and structured, providing a logical progression from motivation through experimental outcomes. The slides were well-organized, and the visualizations were effective in supporting the spoken commentary. However, some pacing inconsistencies occurred during the presentation, likely due to nervousness, which may have led to slightly accelerated explanations in certain segments. While this did not significantly disrupt the audience's comprehension, it may have impacted the emphasis on key findings.

Given that the presentation was collaboratively developed by two individuals, minor discrepancies emerged between the data visualized in some graphs and the spoken interpretations. In particular, there was occasional ambiguity about which dataset corresponded to each chart. This was most evident in the discussion of the "Unserved Customers" metric, which, upon reflection, offered limited analytical value and did not contribute substantially to the interpretation of system performance. Its inclusion in a dedicated graph may have inadvertently diverted attention from more insightful metrics such as wait time and throughput.

Post-presentation, no critical feedback or clarifying questions were offered by the professor regarding the slides or the oral delivery. This absence of notes suggests that, despite minor inconsistencies and pacing issues, the presentation met the expected standards for clarity and content delivery.